Research on Multi-Fidelity Data-Based Optimization Algorithms

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What is multi-fidelity?

- **Fidelity**: how *faithful* a test is
- **High-fidelity**: *expensive*, but precise
- **Low-fidelity**: *cheap*, but a little off
- **Multi-fidelity**: use *both*

How to use them together to find the best answer?

Here's my plan

1. A quick story
2. Two simple ideas
3. What I found
4. Where I'd like to go next

A quick story

Testing is expensive, few chances

You work for a company, but —

- Tune parameters of an *experiment* — every run costs money
- More than 10 parameters, and they *affect each other*
- Can't tune them *one at a time* — need the best combination
- Engines, materials, biology: the same story

How to spend a tiny budget smartly?

Why is it so hard? Two reasons

Two reasons:

- **Black box**: only input and output, never the inside — a simulation or test bench
- **Budget**: an enormous parameter space, but only about 15 runs

Use 15 runs to find the best point of a high-dimensional black box.

Good news: two fidelities

Two fidelities:

- **Low-fidelity**: cheap & fast, but *biased*
- **High-fidelity**: accurate, but only 15 runs

How do we use the cheap low-didelity runs to guide the expensive high-fidelity ones?

My simple idea

My idea is simple:

$$h_{\mathrm{H}}(\boldsymbol{x}) = h_{\mathrm{L}}(\boldsymbol{x}) + \delta(\boldsymbol{x})$$

Learn the correction δ_k from a few high-fidelity runs

Two simple ideas

V. Roshan Joseph

Where to put the first runs?

— **maxpro design**

How to add runs one at a time?

— **sequential design**



V. Roshan Joseph

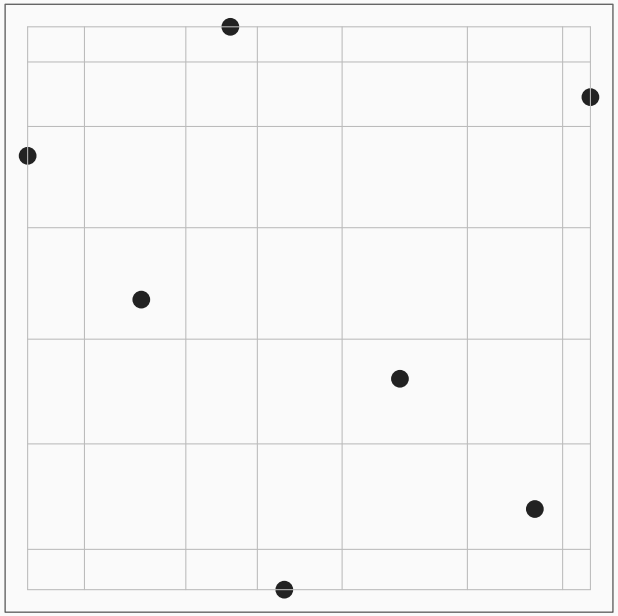
First piece: maxpro design

Maxpro:

maximum projection. spread out in the full space, and in *every low-dimensional shadow.*

Critertion:

$$\min_{D} \sum_{i=1}^{n} \prod_{j \neq i} \frac{1}{\sum_{k=1}^{p} \left(x_{ik} - x_{jk}\right)^2}$$



The MaxPro Design

Second piece: Gaussian progress

- A **normal distribution** describes *one number*: the average, and how spread out
- A **Gaussian process** is the same idea, but for a *whole function*: a whole curve, plus *how unsure* we are at each point
- That *uncertainty* is the guide — test where we are *hopeful and uncertain*

The whole pipeline

1. Choose **starting points** — spread out evenly
2. Feed **many cheap low-fidelity points**; the GP draws a *rough map*
3. Run **one expensive high-fidelity test** where the map says the *destination*
4. **Repeat** — the map gets sharper every round

Map, guess, test, refine.

The formal question

The formal question:

$$\hat{\boldsymbol{x}} \approx \underset{\boldsymbol{x} \in [0,1]^p}{\arg\min}\, h_{\mathrm{H}}(\boldsymbol{x})$$

Find the best point using only **15 high-fidelity tests**.

Three smaller questions:

1. Does **combining the two fidelities** help?
2. Does **screening** help — which parameters matter, drop the rest?
3. Does it still work in **high dimensions**?

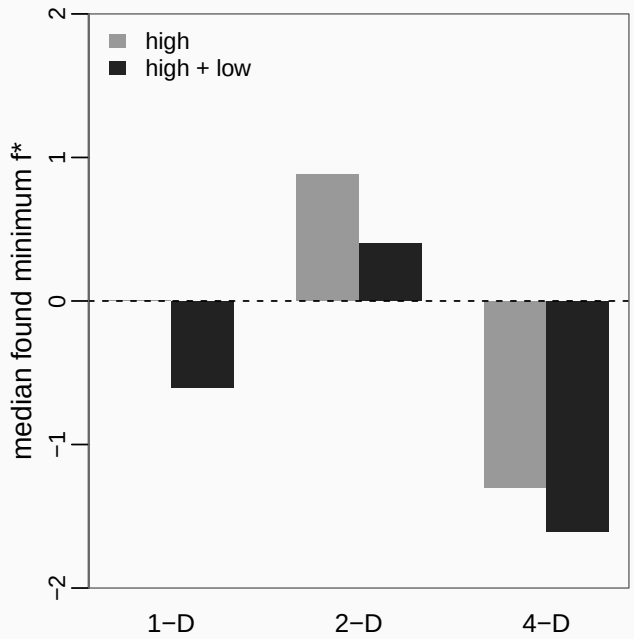
How I measure success

- Run the pipeline **20 times** from scratch
- Take the **median** of the 20 best values
- **Lower is better** in every figure

What I found

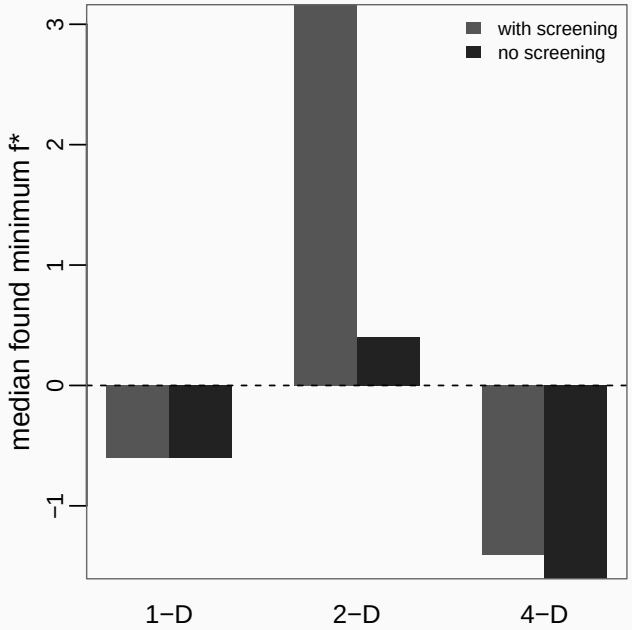
Result 1: combining fidelities is the key

Don't just run the expensive one — **bring a cheap one along to help**.



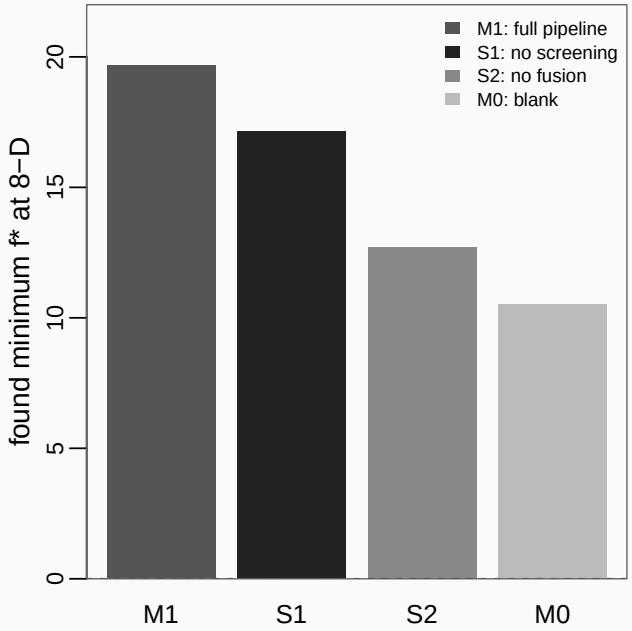
Result 2: screening is useless

Screening sounds like it saves work, but it **costs you accuracy**

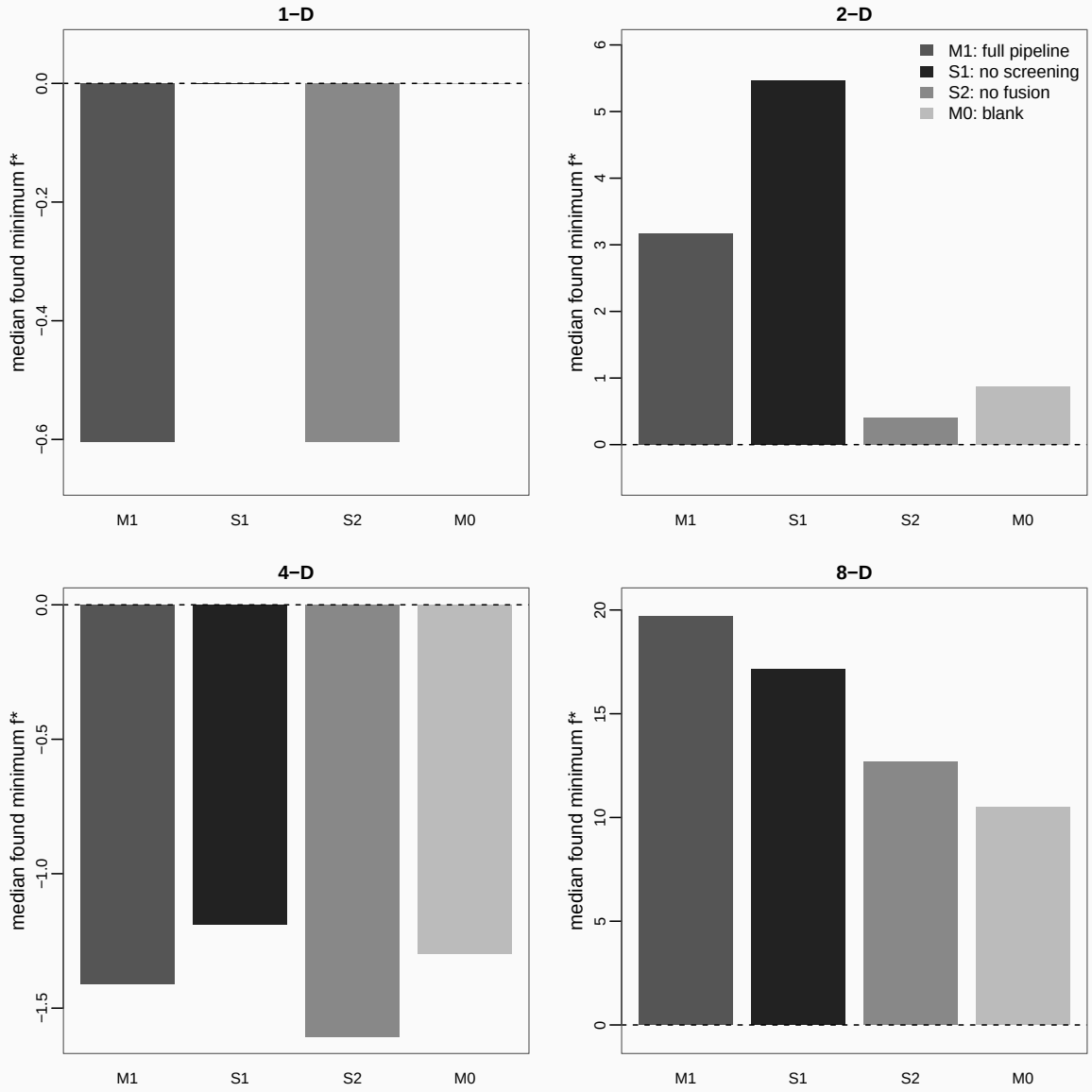


Result 3: high demensions fail

In eight demensions, **hign demensions fail — the curse of demensionality**.



All results in one picture



Where I'd like to go next

Three directions next

1. **When is screening trustworthy?**
2. **Break the curse of dimensionality**
3. **Push it into the real world**

Summary

Don't just **run more high-fidelity tests — make every single one count**. Let cheap low-fidelity runs do the **heavy lifting**.

That's **multi-fidelity**.

Thank you for your listening!